

3.6 Libraries in Python (Built-in & Third-Party)

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے import سے پروگرام میں لایا جاتا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 12 of 13

3.6 Libraries in Python (Built-in & Third-Party)

LEARN IT

A **library** (also called a **module**) is a collection of pre-written code that offers ready-made functions. Instead of writing a square-root algorithm from scratch, you import `math` and call `math.sqrt()`. Libraries save time, reduce bugs and let you stand on other programmers' work.

There are five common ways to import a library. Method 3 (import only what you need) is the best habit: it is clearer and uses less memory than `from math import *`.

Method	Syntax	How you call the function
1. Entire module	<code>import math</code>	<code>math.sqrt(25)</code>
2. Alias (nickname)	<code>import math as m</code>	<code>m.sqrt(16)</code>
3. Specific function	<code>from math import sqrt</code>	<code>sqrt(36)</code>
4. Several functions	<code>from math import sqrt, pi</code>	<code>sqrt(9), pi</code>
5. Everything (*)	<code>from math import *</code>	<code>sqrt(64)</code> — avoid this

WORKED EXAMPLE

Five imports, one result each:

```
import math
print(math.sqrt(25))
import math as m
print(m.sqrt(16))
from math import sqrt
print(sqrt(36))
from math import sqrt, pi
print(pi)
from math import *
print(sqrt(64))
```

OUTPUT

```
5.0
4.0
6.0
3.141592653589793
8.0
```

COMMON MISTAKE

`from math import *` dumps every name into your program. If you later write a function called `sqrt`, you silently overwrite the library one. Prefer methods 1–4.

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے `import` سے پروگرام میں لایا جاتا ہے۔ سب سے اچھا طریقہ یہ ہے کہ صرف وہ فنکشن درآمد کریں جس کی ضرورت ہو، ستارہ `*` سے سب کچھ نہ کھینچیں۔

CHECK YOURSELF

1. Write the line that lets you call `ceil` without writing `math.` in front.
2. Why is an alias such as `import pandas as pd` useful?

Answers: 1. `from math import ceil`. 2. The library name is long; a short nickname keeps later lines short and is a community convention.

LEARN IT

Built-in libraries come pre-installed with Python. You only import them; you do not run `pip`. Three that every Class XI student must know are `math`, `random` and `datetime`.

math

```
import math
print(math.pi)
print(math.ceil(4.3))
print(math.floor(4.9))
print(math.pow(2, 4))
print(math.sqrt(81))
```

OUTPUT

```
3.141592653589793
5
4
16.0
9.0
```

random and datetime

```
import random
print(random.randint(1, 1))
print(0.0 ≤ random.random() < 1.0)

import datetime
now = datetime.datetime(2026, 9, 27, 11, 59, 40)
print(now.strftime("%d-%m-%Y"))
```

OUTPUT

```
1
True
27-09-2026
```

`randint(a, b)` includes both ends. `random()` is a float in `[0.0, 1.0)`. `strftime` formats a date; `%d-%m-%Y` is day-month-year.

REMEMBER IT

`ceil` always rounds **up** (ceiling); `floor` always rounds **down**. `round()` (built-in, not in `math`) goes to the nearest integer and is a different function.

`math` حساب کے لیے، `random` بے ترتیب عدد کے لیے اور `datetime` تاریخ وقت کے لیے بنی ہوئی لائبریریاں ہیں۔ انہیں انسٹال نہیں کرنا پڑتا، صرف `import` کافی ہے۔

CHECK YOURSELF

1. What does `math.ceil(4.0)` return?
2. Write a line that prints a random integer from 1 to 6 (a dice).

Answers: 1. 4 (already an integer). 2. `print(random.randint(1, 6))` after `import random`.

LEARN IT

Third-party libraries are written by the community. They are **not** bundled with Python, so you install them from a terminal with `pip install library_name` and then import them as usual.

Library	Primary use
NumPy	Heavy numerical computing and large arrays
Pandas	Data analysis; table-like data (CSV / Excel)
Matplotlib	Data visualisation — graphs and charts
TensorFlow / PyTorch	Artificial intelligence and machine learning

WORKED EXAMPLE

Installing and importing Pandas (you will use this in Class XII):

```
pip install pandas
import pandas as pd
```

COMMON MISTAKE

`pip` is a **terminal** command, not a Python statement. Do not type `pip install pandas` inside a `.py` file. In VS Code, open the Terminal panel.

اردو میں سمجھیں

تھرڈ پارٹی لائبریریاں پائتھن کے ساتھ نہیں آتیں، انہیں `pip install` سے انسٹال کیا جاتا ہے۔ نمپائی، پینڈا، میٹپلاٹ لیب اور ٹینسر فلو اسی قسم کی ہیں۔

CHECK YOURSELF

1. Which library would you choose to draw a bar chart of class marks?
2. Is `math` a third-party library?

Answers: 1. Matplotlib. 2. No — `math` is built-in.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A library is:

- a. Pre-written functions you import
- b. A logic gate
- c. A cable
- d. A sprint

MCQ 2. math is:

- a. Built-in / standard library
- b. Third-party only
- c. A DBMS
- d. An OSI layer

MCQ 3. pandas is:

- a. Third-party (pip install)
- b. Built into every tiny Python
- c. A sort algorithm
- d. A gate

MCQ 4. import math then square root is:

- a. `math.sqrt(x)`
- b. `sqrt x`
- c. `x**` without import always for sqrt
- d. `print.sqrt`

MCQ 5. `random.randint(1,6)` is for:

- a. A dice integer 1–6
- b. Sorting
- c. Files
- d. TCP

MCQ 6. pip installs:

- a. Third-party packages
- b. The CPU
- c. OSI layer 1
- d. Urdu fonts only

Extra short questions

1. Differentiate built-in and third-party libraries.
2. How do you call a library function?
3. Give two built-in and two third-party examples from this course.
4. Why libraries?

Extra long question

1. Write a tiny program using math and random, then explain how you would add pandas (install + import). Mention a mistake: importing before installing.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Built-in ship with Python (math, random, sqlite3). Third-party need pip (pandas, matplotlib).
2. `import lib; lib.func(...)` or `from lib import func`.
3. Built-in: math, random (or sqlite3 in XII). Third: pandas, matplotlib.
4. Do not reinvent sqrt, CSV readers, or plots.

Long

1. `import math, random; print(math.pi, random.randint(1,6))`. Terminal: `pip install pandas`. Then import pandas as pd. ImportError if pip was skipped.